

## Curated Research Articles

Generated: 2026-06-22 07:20

- **In Situ Characterization Techniques for Investigating the Reaction Mechanisms of Lithium–Sulfur Batteries: Progress, Application, and Future** — score: 1.000 The review article examines the electrochemical challenges faced by lithium-sulfur (Li–S) batteries and highlights the importance of in situ characterization techniques for investigating reaction mechanisms in real-time. By classifying these techniques into three categories—focusing on cathode catalysis, polysulfide shuttling, and anode SEI evolution—the authors propose a systematic framework to guide researchers in selecting appropriate methods to improve the understanding and performance of Li–S batteries, while addressing existing limitations and suggesting strategies for future advancements. Journal: *Wiley: Small Methods: Table of Contents*
- **Tailored Temperature-Responsive Solvation Structures Enabling Ultra-Wide-Temperature Quasi-Solid-State Lithium Metal Batteries From  $-70^{\circ}\text{C}$  to  $60^{\circ}\text{C}$**  — score: 1.000 This study introduces a temperature-responsive gel polymer electrolyte (TRGPE) that optimizes ion coordination with lithium ions, allowing lithium metal batteries to operate stably across a wide temperature range of  $-70^{\circ}\text{C}$  to  $60^{\circ}\text{C}$ . By enhancing low-temperature kinetics and high-temperature stability through unique solvation structures, the TRGPE significantly improves battery performance, evidenced by high discharge capacities in Li//LiCoO<sub>2</sub> cells and the formation of a Li<sub>2</sub>O-rich solid electrolyte interphase. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Operando Full-Field Strain Reconstruction Reveals Geometry-Driven Kinetic Desynchronization in LiFePO<sub>4</sub> Pouch Cells** — score: 1.000 The study utilizes operando optical fiber sensing to analyze the internal strain dynamics of LiFePO<sub>4</sub> pouch cells during fast charging. It identifies that as charging rates increase, the initially synchronous deformation of the electrode transitions to a desynchronized state, revealing localized mechanical heterogeneity and significant strain variation. These findings highlight the importance of internal strain behavior for optimizing battery architecture and improving performance in high-power lithium-ion applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Revealing competitive interfacial reactions in high-energy Li–S batteries** — score: 0.900 The study uses liquid-cell electrochemical transmission electron microscopy to visualize phase segregation in high-concentration interfacial layers of lithium-sulfur batteries. It uncovers competitive interfacial reactions occurring in lean electrolyte environments, providing insights into the complex behaviors at various phase boundaries, which could enhance battery performance. Journal: *Nature*
- **Facet-Driven Interfacial Regulation in a Heterostructured Dual-Host for Highly Reversible Zinc–Iodine Batteries** — score: 0.900 The study presents a novel TiO<sub>2</sub>-coated TiN nanowire array for zinc-iodine batteries, employing a facet-heterointerface approach that significantly enhances battery performance. This design effectively reduces issues of polyiodide shuttling and dendrite formation while promoting uniform zinc deposition through improved interfacial chemistry, resulting in high reversible capacity and excellent cycling stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Molecular orchestration enables mutually reinforcing electrolyte chemistry for lithium metal batteries exceeding 500 Wh kg<sup>−1</sup>** — score: 0.800 The article discusses advancements in lithium metal battery technology achieved through molecular orchestration and complementary charge engineering, which enhance electrolyte chemistry. This approach stabilizes solvation structures and dual interphases, enabling the development of batteries capable of exceeding 500 Wh kg<sup>−1</sup> in energy density. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Entropy-Regulated Dielectric Friction in Eutectic Electrolyte Unlocks  $150^{\circ}\text{C}$  High-Performance Li Metal Batteries** — score: 0.800 This article presents a novel approach to enhance the performance of lithium metal batteries at elevated temperatures (up to  $150^{\circ}\text{C}$ ) by utilizing entropy-regulated dielectric friction in eutectic electrolytes. The research addresses the limitations

of traditional electrolytes, paving the way for more effective use of Li metal batteries in extreme conditions. Journal: *RSC - Energy Environ. Sci. latest articles*

- **Electron-Withdrawing Chemistry Drives Ultrafast-Charging Interphases for Sodium-ion Batteries** — score: 0.800 The article discusses how electron-withdrawing chemistry can enhance the stability and performance of solid electrolyte interphases (SEI) in sodium-ion batteries. By addressing the limitations of conventional carbonate electrolytes, this approach promises to facilitate faster charging and improve the long-term cyclability of these batteries. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Current-Driven Li<sub>2</sub>O Formation in Catalyst-Free Solid-State Li-O<sub>2</sub> Batteries Enabling Simultaneous High Energy and High Power** — score: 0.800 This article presents a breakthrough in solid-state Li-O<sub>2</sub> battery technology, demonstrating a catalyst-free system that effectively enables four-electron Li<sub>2</sub>O formation through a current-driven phase-conversion mechanism. The battery achieves impressive metrics of 1032 Wh · kg<sup>-1</sup> in energy density and 374 W · kg<sup>-1</sup> in power, thus overcoming traditional limitations associated with Li-O<sub>2</sub> batteries and paving the way for next-generation energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Donor–Acceptor Covalent Organic Framework Enables Ambipolar Charge Storage in Aluminum-Ion Energy Storage** — score: 0.800 Researchers have developed a donor–acceptor covalent organic framework (COF) designed to serve as an ambipolar cathode for aluminum-ion energy storage systems (Al-ESS). This innovative COF, characterized by its crystalline and microporous structure, enables efficient charge transport and multi-electron redox reactions, achieving a remarkable specific capacity of 270 mAh/g and excellent cycling stability, thus paving the way for enhanced performance in sustainable energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multidimensional Stress Regulation in Silicon Anodes: From Nanostructures to Electrode Engineering** — score: 0.800 The review article addresses the significant stress challenges faced by silicon anodes in lithium-ion batteries, including mechanical and interfacial stresses, which hinder their performance. It systematically explores various multidimensional engineering strategies—ranging from nanostructural designs to electrode-level improvements—that can help mitigate these stresses and enhance the durability and capacity of silicon-based anodes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Interfacial Chemistry Regulation via AlPO<sub>4</sub> Modification Enables Fast Charging and Wide-Temperature Operation of Graphite Anodes for Lithium-Ion Batteries** — score: 0.800 The article discusses a novel interfacial chemistry regulation approach using AlPO<sub>4</sub> modification to enhance the performance of graphite anodes in lithium-ion batteries. This modification leads to improved reaction kinetics, reduced gas evolution, and a stable, inorganic-rich solid electrolyte interphase (SEI), enabling high-rate capabilities and consistent performance across a wide temperature range from -20°C to 60°C, ensuring longevity and efficiency in battery operation. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Guiding Uniform Nucleation and Growth by Engineered Electrolyte-Philic and Sodiophilic Interface for Dendrite-Free Sodium Metal Batteries** — score: 0.800 The article presents a novel interface made of CH<sub>3</sub>COONa and Na<sub>3</sub>Sb that enhances sodium metal battery performance by promoting uniform sodium nucleation and growth. The combined electrolyte-philicity and sodiophilicity of this interface facilitate improved ionic transport and interfacial wetting, leading to dendrite-free operation and impressive cycle stability across various conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Toward Stable Aluminum Metal Anodes in Nonaqueous Aluminum Batteries: Challenges and Design Strategies** — score: 0.800 The article examines the stability issues of aluminum metal anodes in nonaqueous aluminum batteries, highlighting that their failure is due to interconnected degradation processes rather than isolated defects. It categorizes stabilization strategies into four key areas: anode optimization, interfacial engineering, electrolyte enhancement, and the design of solid/gel/quasi-solid electrolytes, emphasizing the need for a coordinated approach to enhance anode

performance and durability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Decoupling polarization in extreme fast-charging LiFePO<sub>4</sub> batteries: from mechanisms to rational design** — score: 0.800 The article discusses the polarization challenges in LiFePO<sub>4</sub> batteries during extreme fast-charging and explores the underlying mechanisms contributing to this issue. It aims to inform the rational design of these batteries to enhance their performance and efficiency by minimizing polarization effects. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Linearly Bonded Organoselenide-Sulfide Hybrid Compound Cathode Materials for High-Capacity and Shuttling-Suppressed Lithium-Organochalcogenide Batteries** — score: 0.800 The article presents a novel organoselenide-sulfide hybrid cathode material that enhances the performance of lithium-organochalcogenide batteries by minimizing shuttling effects and increasing electrochemical efficiency. Its unique chain-like structure and high atom ratios improve Li<sup>+</sup> diffusion, resulting in a specific capacity of 816.8 mAh/g and excellent capacity retention, paving the way for practical applications in high-performance battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A Bioinspired CNTs-Enhanced Anode for Ultra-Stable Aluminum-Ion Batteries** — score: 0.800 This article presents a novel bioinspired aluminum ion battery (AIB) anode enhanced with carbon nanotubes (CNTs) that mimics the nerve net of sea anemones. By creating a dynamic 3D network, the anode effectively manages charge distribution and prevents dendrite formation, achieving over 1800 hours of stable cycling and significantly outperforming traditional aluminum anodes in both stability and capacity retention. This advancement offers a promising approach for enhancing energy storage systems in practical applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Weakly Coordinating Co-Solvent Design of Chlorine-Free Electrolytes: Manipulating Solvation Structure and Interfacial Chemistry for Highly Reversible Mg Metal Anodes** — score: 0.600 The article discusses the design of chlorine-free electrolytes that utilize weakly coordinating co-solvents to enhance the solvation structure and interfacial chemistry of magnesium metal anodes. This approach aims to achieve highly reversible performance in magnesium batteries, addressing the challenges posed by existing electrolyte formulations. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Synergistic Weak Solvation and Electrostatic Shielding Enabled by Ammonium Hypophosphite Additive for Fast and Stable Zn Storage** — score: 0.500 The article discusses a novel approach to enhance zinc storage performance by using ammonium hypophosphite as an additive, which promotes synergistic weak solvation and electrostatic shielding. This strategy results in faster and more stable electrochemical reactions, thereby improving the overall efficiency and longevity of zinc-based storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Conducting-Polymer-Coated Nanocellulose Enabling Dual-Gradient Thick Electrodes for Redox-Homogeneous Ultrahigh-Areal-Capacity Li-Ion Batteries** — score: 0.400 The article discusses the development of dual-gradient thick electrodes using conducting-polymer-coated nanocellulose for lithium-ion batteries. This innovative design addresses the challenge of redox heterogeneity, which is typically caused by material segregation in thick electrodes, thereby improving energy density and performance in battery applications. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Advanced Materials for Seawater Batteries toward Blue Energy Storage: Key Strategies and Future Perspectives** — score: 0.400 This review addresses the challenges and strategies involved in developing seawater batteries (SWBs), emphasizing the importance of seawater chemistry in determining battery configurations and material needs. It discusses design approaches to enhance electrochemical performance through better electron/ion transport and interfacial control, outlines potential applications, and identifies future research directions for creating advanced materials aimed at improving SWB efficiency and longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Inhibition of shuttle effect in organic cathode and in situ mechanistic exploration** — score: 0.400 The article explores the inhibition of the shuttle effect in organic p-type cathode materials, which

are advantageous for energy storage because of their sustainability and high voltage capabilities. The authors investigate the underlying mechanisms of this inhibition through in situ methods, potentially enhancing the performance and applicability of these materials in energy storage systems. Journal: *RSC - Energy Environ. Sci. latest articles*

- **Lewis acidic molecule-modulated synchronous anion immobilization and decomposition towards a robust solid electrolyte interphase with enhanced Zn<sup>2+</sup> kinetics** — score: 0.400 The article discusses a novel approach to enhancing zinc-ion batteries by utilizing Lewis acidic molecules to regulate anion immobilization and decomposition processes, ultimately leading to a more stable solid electrolyte interface (SEI). This method aims to mitigate issues such as dendrite growth and side reactions in zinc metal anodes, thereby improving battery performance and longevity. Journal: *RSC - Energy Environ. Sci. latest articles*
- **The Behaviour of Phenothiazines as Catholytes in Aqueous-Organic Redox Flow Batteries** — score: 0.400 The study investigates various commercial phenothiazine dyes as potential catholytes in aqueous-organic redox flow batteries. Among the tested dyes, Azure-A emerged as the most effective candidate, suggesting its potential for enhancing battery performance. Journal: *RSC - EES Batteries latest articles*
- **Nanoscopic Observation of Structural Changes After First-Firing Behavior in W/GeTe/W Ovonic Threshold Switches Using AFM Tip-Based Local Biasing** — score: 0.400 The study utilizes atomic force microscopy with local biasing to observe structural changes in GeTe films during Ovonic threshold switching, revealing the formation of nanocrystals at the electrode interface. First-principles calculations highlight that these changes lead to mid-gap states and enhanced charge redistribution, significantly lowering the voltage needed for subsequent switching after the device's initial operation. Journal: *Wiley: Small Methods: Table of Contents*
- **Thermal Gradients Increase Concentration Polarization Resistance and Reduce Lithium Diffusion in NMC/Graphite Batteries** — score: 0.400 The article investigates how thermal gradients in NMC/graphite batteries enhance concentration polarization resistance while simultaneously limiting lithium diffusion. This phenomenon has implications for the efficiency and performance of energy storage systems, indicating potential challenges in managing thermal conditions within batteries. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Frustrated polar slush-like state leads to advanced stability and high energy storage in NaNbO<sub>3</sub>-based ceramics** — score: 0.300 The article discusses the discovery of a frustrated polar slush-like state in NaNbO<sub>3</sub>-based ceramics, which enhances their stability and energy storage capabilities. This advanced material could have significant implications for the development of high-performance energy storage systems. Journal: *ScienceDirect Publication: Nano Energy*
- **Inline quality grading of commercial lithium-ion battery manufacturing via data-efficient learning and transferable assessment** — score: 0.200 The article discusses the importance of inline quality grading in the manufacturing of lithium-ion batteries, emphasizing the need for reliable methods to ensure consistency and efficiency. It highlights the use of data-efficient learning techniques and transferable assessment methods to improve existing grading paradigms in this complex manufacturing process. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Fast formation to reinforce lithium-rich cathodes** — score: 0.200 The article presents a novel fast formation process for lithium-ion batteries that significantly enhances their performance compared to traditional slow formation methods. This approach not only reduces costs but also boosts capacity, stability, and cycle life, making it applicable to a wider range of electrode systems, particularly lithium-rich cathodes. Journal: *Nature*
- **Non-corrosive Difluoromethylene-Enriched Sulfonimide Lithium Salt Electrolyte for 4.6V Lithium Metal Batteries** — score: 0.200 The article presents a novel non-corrosive electrolyte consisting of difluoromethylene-enriched sulfonimide lithium salts designed for use in 4.6V lithium metal batteries. This research potentially addresses stability and performance issues in high-voltage applications, contributing to the advancement of battery technology. Journal: *ScienceDirect Publication:*

- **Design of Experiments Guided Identification of Coupled Component Interactions in Cu Gas Diffusion Electrodes for Electrochemical CO<sub>2</sub> Reduction to Multi-Carbon Products** — score: 0.200 This study employs a Design of Experiments (DoE) methodology to investigate the interactions of five key components in Cu gas diffusion electrodes (GDEs) for electrochemical CO<sub>2</sub> reduction, revealing how component variations influence product yields. By utilizing a definitive screening design (DSD) framework, the researchers identify critical performance factors, particularly the roles of Nafion and carbon, and validate their findings experimentally, achieving significant enhancements in C<sub>2</sub>H<sub>4</sub> selectivity through optimized carbon loading. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Main-Chain Ion-Pair Polybenzimidazole Membranes Enabling Reduced-Temperature HT-PEMFC Operation (Down to 120°C) (Adv. Energy Mater. 23/2026)** — score: 0.200 The article discusses a novel polybenzimidazole membrane infused with imidazolium–biphosphate ion pairs, which effectively retains phosphoric acid and enhances proton conduction. This innovative design not only minimizes acid loss but also promotes high conductivity and stability, enabling high-temperature proton exchange membrane fuel cells to operate efficiently at temperatures as low as 120°C. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Gas Immersion Enabled Atmospheric-Pressure Near-Ambient Phase-Change Refrigeration Cycle for Active Safety Protection of LFP Battery Thermal Runaway: A Single-Cell Proof of Concept** — score: 0.200 This article presents a novel refrigeration cycle using gas immersion to mitigate fire risks from thermal runaway in lithium iron phosphate (LFP) batteries. The study explores a proof of concept for employing this technology at atmospheric pressure to enhance safety by controlling the venting gases that can ignite during battery failure. Journal: *RSC - EES Batteries latest articles*
- **Operando microscopy for neuromorphic hardware** — score: 0.200 This review discusses advanced microscopy techniques—ranging from electron and X-ray to optical methods—aimed at investigating the dynamic behaviors of neuromorphic materials. It emphasizes the significance of operando characterization in understanding the performance of neuromorphic devices in real-time. Journal: *Nature Materials*
- **Homo-Electronic-Type Inner- and Outer-Wall Double-Wall Carbon Nanotubes for Record-Efficiency Carbon/Silicon Heterojunction Solar Cells** — score: 0.000 This study clarifies the long-standing debate on the performance of semiconducting versus metallic double-wall carbon nanotubes (DWCNTs) in carbon nanotube-silicon heterojunctions by comparing high-purity homo-electronic types. The researchers achieved record solar cell efficiencies exceeding 22% and demonstrated that electronic uniformity plays a crucial role in enhancing performance, providing insights that could improve future solar cell designs, including perovskite and tandem systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Ordered Lattice at Polycrystalline-Single Crystal Island Interfaces for Efficient CO<sub>2</sub>/H<sub>2</sub>O Co-Electrolysis in Solid Oxide Electrolysis Cells** — score: 0.000 The study presents an innovative approach to enhance the efficiency of CO<sub>2</sub>/H<sub>2</sub>O co-electrolysis in solid oxide electrolysis cells by constructing a polycrystalline-single crystal island homojunction through epitaxial growth. This design minimizes charge transfer resistance and ensures effective spatial separation of CO<sub>2</sub> and H<sub>2</sub>O adsorption sites, achieving a faradaic efficiency of 95.9% with stable operation over 300 hours while allowing for adjustable syngas production ratios. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Nanocrystal-tailored recombination for all-perovskite tandem solar modules** — score: 0.000 The article discusses advancements in all-perovskite tandem solar modules through the use of nanocrystal engineering to enhance charge recombination processes. This innovative approach aims to improve the efficiency and stability of these solar modules, potentially accelerating their adoption in renewable energy applications. Journal: *Nature*

- **The latest benefit of obesity drugs: boosting testosterone and sperm quality** — score: 0.000 A preliminary study has revealed that GLP-1 obesity medications not only aid in weight management but also improve testosterone levels and sperm quality in users. These findings suggest potential reproductive health benefits for individuals undergoing treatment with these drugs. Journal: *Nature*
- **Why people should work together to shape the economy** — score: 0.000 The article discusses the importance of community engagement in shaping economic policies rather than passively accepting their outcomes. It emphasizes that collective action among individuals can empower communities to influence economic decisions, ultimately benefiting their local environments and economies. Journal: *Nature*
- **At-home brain implant gives man with motor neuron disease his daily life back** — score: 0.000 A novel at-home brain implant has significantly improved the quality of life for a man suffering from motor neuron disease by enabling him to communicate and operate his computer independently for nearly two years. This advancement showcases the potential of neural devices in assisting individuals with severe mobility and communication challenges. Journal: *Nature*
- **How AI is revealing the secret lives of animals from hummingbirds to pumas** — score: 0.000 Recent advancements in machine learning and technology are enabling researchers to track and analyze the behaviors and interactions of various animal species, from hummingbirds to pumas, providing new insights into their movements and social structures. This innovative approach is enhancing our understanding of wildlife ecology and behavior. Journal: *Nature*
- **In the field — and sometimes in the lab — the simplest tool is best** — score: 0.000 The article discusses how the integration of simple, everyday tools into research methodologies can enhance reproducibility and accessibility in scientific studies, despite the prevalence of advanced technological innovations. It highlights the value of simplicity in both field and lab settings to improve research outcomes. Journal: *Nature*
- **Rethinking the green power grid for stability, not just for climate** — score: 0.000 The commentary addresses the 2025 Iberian blackout, highlighting the need to rethink renewable power grids for enhanced stability alongside environmental benefits. It suggests that integrating advanced technologies like synthetic inertia and improved inverter controls, coupled with strategic transmission planning and energy storage advancements, can bolster grid resilience while supporting decarbonization efforts. Journal: *Joule*
- **2D perovskite as three-in-one interface modifier for efficient Pb-Sn perovskite solar cells and all-perovskite tandems** — score: 0.000 The study introduces a 2D perovskite as a multifunctional interface modifier that enhances the quality of Pb-Sn perovskite films, significantly improving solar cell performance. This approach results in Pb-Sn perovskite solar cells achieving an efficiency of 24.07% and enables a two-terminal all-perovskite tandem device to reach a certified efficiency of 29.22%, demonstrating both operational stability and a competitive shelf-life. Journal: *Joule*
- **Efficient organic photovoltaics with thermalization-controlled exciton management** — score: 0.000 The study introduces a novel TADF material, BPhA-QAO, designed to enhance organic photovoltaic (OPV) efficiency by managing exciton dynamics. By optimizing the energy levels of polymer donors, it increases singlet exciton lifetimes and reduces thermalization losses, achieving a certified power conversion efficiency of 20.18% and presenting a strategic approach for advancing OPV performance. Journal: *Joule*
- **Author Correction: UV and thermally stable hole-selective contacts with enhanced assembly density for inverted perovskite solar cells** — score: 0.000 This article presents a correction to previous findings related to hole-selective contacts in inverted perovskite solar cells, highlighting improvements in stability against UV and thermal exposure, as well as increased assembly density. These advancements aim to enhance the efficiency and durability of perovskite solar technologies. Journal: *Nature Materials*
- **Material considerations of memristors at the nanoscale** — score: 0.000 The article features an

interview with Ilia Valov, who explores how nanoscale structural and compositional factors influence the electrochemical behavior of memristors, highlighting their impact on the devices' properties and functionalities. Journal: *Nature Materials*

- **Excitonic spin torque in a magnetic semiconductor** — score: 0.000 The study focuses on the two-dimensional antiferromagnetic semiconductor CrSBr, revealing unique magnon dynamics that emerge at elevated excitation pump power. This behavior is linked to the interaction between excitons and the material's magnetization, suggesting potential applications in spintronics and magnetic information processing. Journal: *Nature Materials*
- **Chain entanglements enable regeneration of high-performance thermosets** — score: 0.000 The study reveals that thermoset materials can achieve high mechanical performance through chain entanglements rather than solely relying on crosslink density, which traditionally limits their recyclability. This innovation allows for the regeneration and reprocessing of glassy thermosets while maintaining their load-bearing capabilities, offering a significant advancement in material sustainability. Journal: *Nature Materials*
- **Fully coupled mechano-electrochemical modeling of 3D-resolved microstructures for solid state batteries manufactured by wet processing** — score: 0.000 The article presents a comprehensive mechano-electrochemical model focusing on the 3D microstructures of solid-state batteries produced through wet processing methods. This modeling approach aims to deepen the understanding of battery performance by integrating mechanical and electrochemical phenomena at a microscopic level. Journal: *ScienceDirect Publication: Energy Storage Materials*